12.13 The Truel

WORKSHEET

The question

STEP II 1995, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself In a two-person duel, Percy throws first. Either he hits at once, or both miss and the situation repeats exactly. Write an equation for Percy's probability of winning.

STAGE 2 Algernon's three options

Ask yourself In each case, who is left, and who throws next?

STAGE 3 Advice

Ask yourself If Algernon aims at Cuthbert, he might hit (probability $\frac{2}{5}$) or miss. What is his overall chance? Compare with deliberately missing.

What's really going on

Strategy and game theory

The truel shows the central idea of *game theory*: the best action depends on how the other players will respond. Hitting Cuthbert, the most dangerous opponent, is the obvious move, but it changes the game in a way that hurts Algernon. In this question Bertie's and Cuthbert's strategies are fixed (they aim at each other); in a full game-theoretic analysis each player would choose their own strategy, knowing that the others are doing the same. Such analyses, starting with the work of John von Neumann and Oskar Morgenstern in 1944 and John Nash in 1950, now underpin economics, evolutionary biology and political science.

Strength as a weakness

Who wins the truel? With Algernon's best plan, the probabilities are $\frac{226}{475}$ for Algernon, $\frac{135}{475}$ for Bertie and $\frac{114}{475}$ for Cuthbert. The best shot is the least likely to survive, because everyone with a choice aims at him. The same pattern appears in elections with several candidates, in competition between firms, and in nature, where the strongest competitor can be the first target. Martin Shubik, who introduced truels into game theory in the 1950s, used them to ask whether the fittest necessarily survive.

Try this next

1 **Bertie and Cuthbert.** Verify that, with Algernon missing deliberately, Bertie wins with probability $\frac{135}{475}$ and Cuthbert with $\frac{114}{475}$.

2 **A better shot.** If Algernon hit with probability a instead of $\frac{2}{5}$, for which values of a is deliberately missing still his best first move?

3 **Should Bertie aim at Cuthbert?** Given that Algernon misses deliberately, is it in Bertie's interest to aim at Cuthbert rather than at Algernon? (Assume Cuthbert's behaviour is unchanged.)

4 **A duel with a fixed number of throws.** In a duel where each player has only two throws, find Percy's probability of winning, and compare with the unlimited duel.

Further reading

- **D. Marc Kilgour & Steven Brams, "The truel"**, *Mathematics Magazine* 70 (1997). An accessible survey of three-way duels.
- **Avinash Dixit & Barry Nalebuff, *Thinking Strategically*** (Norton). Game theory for a general reader, including the truel.
- **Ken Binmore, *Game Theory: A Very Short Introduction*** (Oxford University Press).

The full guided solution is in Section 12.13:
https://beyondstep.keithmcnulty.org/ch12/truel.html